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1174: FUTURISTIC TRENDS AND INNOVATIONS IN MULTIMEDIA SYSTEMS USING BIG DATA, IOT AND CLOUD TECHNOLOGIES (FTIMS)

Fiber Bragg grating sensors driven structural health monitoring by using multimedia-enabled iot and big data technology

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Abstract
Structural Health Monitoring (SHM) of large structures is a critical aspect due to various environmental conditions, high speed & long-distance communication, dynamic analysis of the structure, and cost of operation. These issues can be addressed using Fiber Bragg Grating (FBG) sensor technology which has evolved to a new height and is widely used in various distributed critical sensing applications. These are mostly preferred due to long-distance monitoring, low cost of operation, and immunity to Electromagnetic (EM) radiations. Similarly, the monitoring of a large structure from a long distance is also one of the crucial aspects of SHM technologies. These technological challenges can be addressed using an integrated distributed sensing solution consisting of FBG sensors, Big Data, Kafka, and the Internet of Things (IoT). In this article, the fabrication of the FBG sensor and the bonding of the sensing element to the base plate of the suspension bridge structure are discussed along with experimental details. A scalable architecture of the proposed Smart Distributed Sensing (SDS) model using FBG sensors is also discussed in this article. The experimental validation is performed using an IoT based FBG sensing mechanism to estimate the strain distribution profile at the bonding region of the base plate from a central location.

Keywords FBG · Sensor · Sensor calibration · Sensor signal processing · Strain distribution analysis · Structural health monitoring · Critical sensing · Big data · IoT · Kafka · Smart distributed sensing

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1 Introduction

FBGs are utilized as passive optical sensors for different purposes like SHM, biomedical sensors, sensors in aviation applications where EM radiation may introduce variations during real-time measurements when using conventional sensors. These sensors can be utilized to gauge different measurements like strain, displacement, acceleration, vibration and tilt angle, and so forth. These FBG sensors are sensitive to the impact of strain on the Fiber Grating region. Different transduction principles or mechanisms are utilized to convert the physical sensing parameter to a strain change on the Fiber Grating region [27]. These FBG sensors are fit to detect extremely small strain and temperature change, with strain sensitivity of 1.2 pm/μstrain and 11 pm/°C respectively. The critical aspect of the FBG sensing scheme is the calibration and compensation technique, which is followed in every installation area.

The commercial FBG sensors have strain and temperature sensitivity of 1.2 pm/μstrain and 11 pm/°C respectively [18, 28, 31, 35]. The FBG sensors are most suitable for structural dynamic analysis which covers the impact of dynamic loading and the effect of high acceleration on a structure. The dynamic loading includes various other elements such as wind speed, the impact of waves, traffic density, vibration due to earthquake and blast, etc. The dynamic analysis gives a wide perspective for dynamic displacement analysis, maintaining time history, and performing modal analysis.

Further, an FBG based sensor offers various other advantages over other electronic sensors such as immunity to EM radiation, high sensitivity, multiplexing as well as distributed sensing capacity [4, 22]. The dynamic analysis of any structure requires a large number of measurement samples to collect adequate structural parameters. Therefore, the high-speed sensing and monitoring mechanism requires a sensing scheme which can measure at a strain and temperature sensitivity of 1.2 pm/μstrain and 11 pm/°C respectively. The limit of sensitivity may change depending on the precise measurement of dynamic parameters like the speed of response, fidelity, natural frequency, and location of the applied load, etc. The possible solution to handle such large volumes and velocity of data is the IoT and distributed sensing. The conventional sensors are not suitable for the measurement of dynamic parameters at a speed of more than 2.5 kHz with a low-cost solution like Fiber Optics. In addition to the above aspects, current IoT systems do not require such high-speed measurement. In a similar context, the Internet of Things (IoT) based distributed sensing approach will provide better scalability as compared to an offline monitoring mechanism [12, 23, 26]. Thereby, FBG sensing systems can be integrated into an IoT framework for better signal processing, historical data management, scalable solutions, and smart Decision Support along with notification generations. These systems have added advantages to provide a complete Monitoring, Processing and Notification (MPN) approach where cross-platform service is not an issue. The major benefit is to connect a large volume of users to the proposed sensing system through smart devices like Android/IOS devices and tablets. The research article focuses on the two major objectives to solve issues involved in sensing, data acquisition, and processing in real-time. The primary objective of this work is to develop a sensing mechanism for the acquisition of sensor signal at a high-speed mode and the secondary objective is to sense the strain profile of a region on the base plate of the suspension bridge model. The soft sensor technology is used to estimate the complete strain profile of a region on any structure. A single FBG sensor is used to evaluate the strain profile of a region on the base plate. Apart from the above aspects, the signal processing of the sensor has a major role in extracting useful information of any structure. The real-time

FBG sensor signal contains a noise level which is also required to be removed from the original signal. All these technological aspects are discussed in the article with experimental validation.

This article comprises of seven sections. The second section "Literature Review" discusses various technological advancements of FBG sensors in SHM and IoT. The complete details of the experimental setup are discussed in the third section "Experimental Setup". The fourth section of the article portrays the methodology adopted to conduct the experimental study followed by the results. The conclusion section of the article discusses the precise decisions reached during the experimental analysis with validated mathematical models.

2 Literature review

Presently, there are various types of optical sensors and optical sensing schemes are available for the measurement of several physical quantities such as displacement, velocity, tilt angle, cracks, and pressure, etc. Several research articles are published on the architecture of remote measurement schemes in SHM applications. The usage of FBG sensors and connecting FGB sensing systems to IoT environments in SHM applications are not much explored. Similarly, IoT is an emerging field of research as data velocity, data variety, data volume, and data securities are the major concerns of all researchers. The FBG sensing mechanisms are not cost-effective solutions in every application but it is mostly preferred as the speed of measurement, site maintenance, and dynamic analysis of structures which has been a perennial problem for the industries, is relatively easily dealt with using FBG sensors. Rekha et al. have proposed WSN (Wireless Sensor Network) based SHM framework and the decision support mechanism to generate adequate notifications for the authorities. The security aspect of data security mechanisms is not discussed in the article. Apart from this, the speed of monitoring and acquisition of real-time data is quite less as compared to fiber optics-based sensing systems [29]. Similarly, the article published by Shigeru et al. portrayed Structural Integrity F-test (SI–F) for the monitoring of fatigue crack in a structure. The strain distribution was also evaluated using Finite Element Analysis (FEA) of the same structure. The proposed system shows a higher sensitivity as compared to visual inspection of any structure [32]. Amir et al. have presented a detailed literature review on smartphone-based monitoring of civil structures. In a similar context, a case study on the deployment of smart devices in the civil structures is presented in a detailed manner. Historical data management is also discussed in the article [2].

Zhang et al. presented a particle swarm optimization support vector regression to achieve self-repairing of FBG sensor network in SHM applications. Eight-point FBG sensors are installed in the wing box of aircraft and the structural analysis is performed. The application of the non-reconstruction based PSO-SVR model is found to be a better approach as compared to simple non-reconstruction models [34]. Further, Jin et al. have investigated the performance of anodized and uniform FBG sensors in a structural crack propagation monitoring application. The reflection spectra under crack generated in non-uniform strain were studied using both theoretically and experimentally. In this research article, three spectral indices central wavelength shift, a ratio of side lobe to the main lobe, and information entropy were investigated to estimate the crack propagation [11].

Similarly, Ruiya et al. proposed a theoretical sensing scheme using FEA and experimental test cases to validate the theoretical model. Two FBG sensors are developed with

an overall sensitivity of 7.72 pm/με and − 2.94 pm/με respectively. The overall sensitivities are 6.43 and − 2.45 times of the strain sensitivity of a bare FBG sensor. These sensors are most suitable in varying temperature environments for critical mechanical equipment [16]. Furthermore, many research reports have been published on the FBG sensing mechanism in SHM applications. There are various kinds of research works published until now; some studies are based on the development of higher accuracy-based FBG sensors, calibration models for FBG sensors, localization of fractures, and development of novel sensors for pressure, temperature, soil strength, air particle monitoring. Few research articles are listed in Table 1 which depicts the novelty approach of the proposed sensing mechanism.

The generic view of the SHM is portrayed in Fig. 1 where various technological aspects are shown. The FBG sensor-based SHM application comes in the first category of the list shown in Fig. 1. The above review is performed to gather parametric performance evaluation of various SHM applications and technological comparisons.

There is a need of such IoT enabled distributed monitoring of civil structures to remotely monitor health conditions of the structures. There are various challenges to be looked after before implementing such distributed monitoring techniques. It is observed from the review that various types of technologies are used to be applied in the SHM applications such as WSN, IoT, and soft approaches. However, the real-time signal from the FBG sensing system (Interrogator) is badly affected by noise and other aspects like distortions. The signal filtering and calibration of the FBG sensor are not explicitly discussed in any article. The calibration mechanism is one of the important aspects of any sensing system. Therefore, signal filtering mechanism, calibration model, IoT based monitoring scheme, Big Data-based large data handling, and decision support mechanism [7, 14, 24] are chosen to discuss in this article. Apart from that, the dynamic analysis of any structure is also possible using this FBG technology which requires multimedia enabled IoT and BigData platform for large volume of data analysis. In addition to the above studies, the characteristic analysis of the FBG sensor for real-time computation via a Big Data platform and use case of Kafka connect together with the strain calibration scheme are outlined in this article. The individual aspects such as Real-Time FBG Signal Filtering, Sensor Calibration, Sensor Signal Processing (SSP), Strain Distribution Analysis, and the real-time system implementation technique are communicated in this article.

3 Experimental setup

An experimental model of a prototype suspension bridge is designed using a CAD software "CATIA". The CAD model is fabricated using gray forged iron material having a tensile strength of 180 MPa at the Laboratory. The stress, strain, and maximum deformation of the suspension bridge are studied using the ANSYS Finite Element Analysis (FEA) model. Firstly, the FBG sensor is bonded on the base plate of the suspension bridge model as shown in Fig. 2. The sensor is interfaced with Ibsen IMON 512 High-speed FBG interrogator, Ibsen SLED light source, and the signal acquisition is performed via LAN interface with the host PC. Further, the wavelength shift is recorded using IMON 512 FBG Interrogator (1528 nm–1568 nm, 2.5 kHz) to develop the load calibration model. The data acquisition is performed using the LabVIEW DLL library file supplied by the vendor. A LabVIEW application is developed to acquire the real-time signal from the FBG sensors and to perform the pre-processing operations on the raw signal. The LabVIEW DLL helps to initialize the connectivity with the FBG

Table 1 Research highlighted on FBG sensor in SHM applications

Author	Methodology	Novelty
Yu et al. (2018)	Low-Velocity Impact (LVI) monitoring mechanism is proposed which is based on Detrended Fluctuation Analysis (DFA) and Centroid Localization algorithm [33]	The average error of localization was 31.5 mm
Magdalena et al. (2017)	SHM was implemented on a wind turbine model to detect the damage and localize it [20]	Identified the location of the damage
Li et al. (2019)	Fiber Bragg grating (FBG) demodulators based on tunable lasers was proposed for SHM applications. Sensor accuracy was improved [17]	FBG interrogation accuracy was raised from ± 3 pm to ± 1 pm
Tahar et al. (2017)	Fiber Bragg Grating Sensors (FBGS) are based on wireless sensor network (WSN) [7]	An efficient method to measure the variation of strain, temperature, and humidity (SV, ST, SH) by modeling the spectrum of FIGS

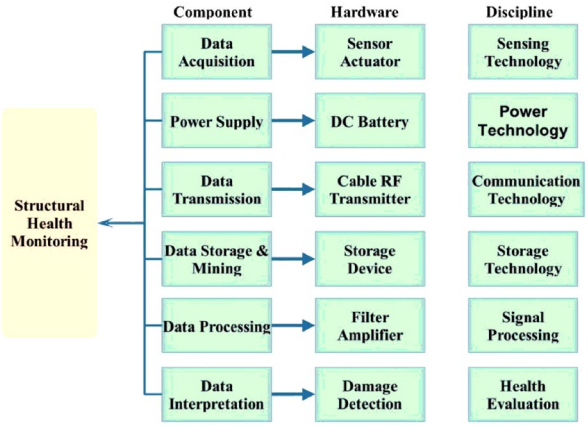


Fig. 1 Generic technological aspects of SHM applications

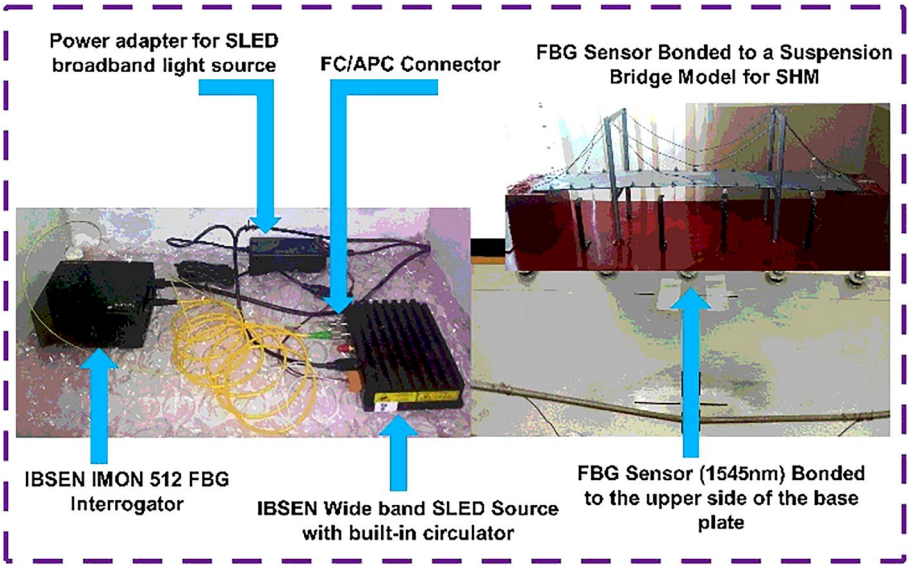


Fig. 2 Experimental setup of the prototype suspension bridge structure bonded with FBG Sensor

interrogator for acquiring the raw signal. The pre-processing of the raw signal includes the noise cancellation model, sensor calibration model and modules to forward the processed information to the Hadoop Distributed File System (HDFS). This pre-processing operation is handled inside a Primary Load Calibration (PLC) module developed using LabVIEW environment. The PLC module is tested in a real-time approach to generating appropriate notifications. The PLC module contains various Sensor Signal Acquisition (SSA), Sensor Signal Processing (SSP) elements, and Signal Measurement (SM) elements.

The SSA elements include the vendor-specific driver file (DLL file) and the callback function implemented to send the pre-processed information to a single node Hadoop cluster. The SSP element contains some of the major functionalities like Down Sampling (DS) and Moving Average Filter (MAF). Similarly, the SM elements include the measurement

functionalities like Signal Peak Detection (SPD), Estimation of wavelength shift, and Stress–Strain (Stress and Strain) Profile, etc.

The SM element also performs the interface to connect Hadoop single node cluster using Kafka packets and forwards the pre-processed information to the HDFS. The Decision Support System (DSS) is developed using Python scripts to import the pre-processed information from the HDFS and to generate adequate notifications which is again forwarded to an open source IoT dashboard (ThingSpeak). The single-node cluster is set up on a Computer (Intel® Core™ i5-9500 T (6 Cores/9 MB/6 T/2.2 GHz to 3.7 GHz/35 W), 8 GB, 1X8 GB, DDR4 non-ECC Memory, 2.5 inches 1 TB 7200 rpm SATA Hard Disk Drive). The complete architecture of the experimental setup and IoT architecture is shown in Fig. 3. The methodology adopted to evaluate the complete work is discussed in the fourth section of the article. Further, the FBG signal is extracted from the sensor via Ibsen IMON FBG interrogator into the host computer using Ethernet LAN. The extraction of data by the host computer is done by the LabView DLL file given by the vendor. Raw data first goes into the DLL file which generates FBG signal data. The signal from FBG sensor is wavelength specific and it is a time-domain signal [25]. This signal is processed locally and is also sent to Local storage infrastructure via Ethernet using the MQTT protocol. The publisher on the host computer publishes FBG signal data to an MQTT topic name "FBG Signal Data".

MQTT has its advantage like, it is lightweight, provides low latency, and supports a large number of client connections at a given instant [8, 10]. For this purpose, Confluent's MQTT Kafka connector is used which pulls the data from one of the topics in the MQTT broker and stores it in one of the topics in Kafka [3, 6]. This brings in all the advantages of Kafka in terms of scalability, persistence, and throughput [5, 13].

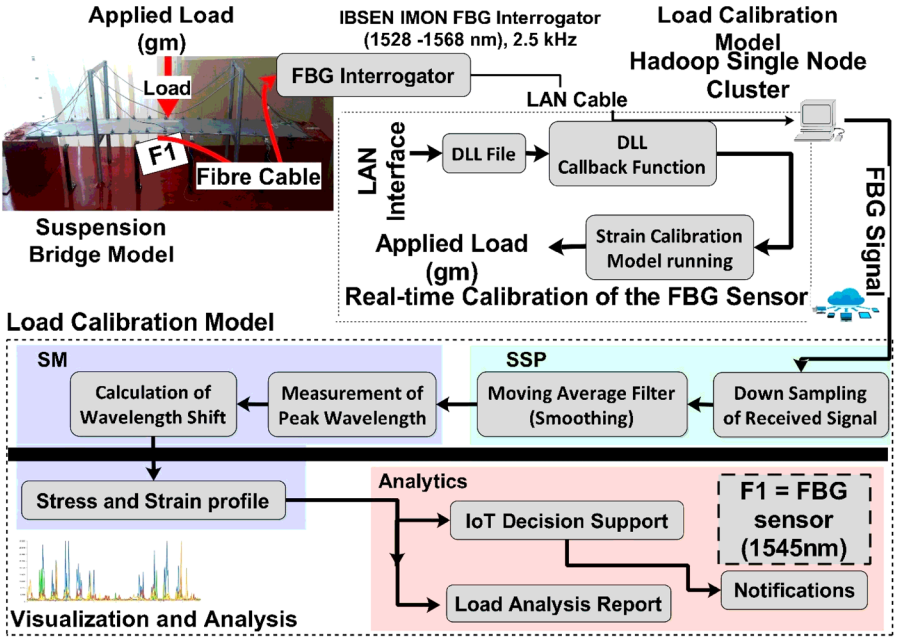


Fig. 3 Block diagram of the experimental setup and IoT architecture

The system testing is performed using a single FBG sensor interrogation channel and MQTT Kafka connects in the single node cluster computer. Hence, all the data was stored in a single Kafka topic. A Kafka-console consumer was run to consume messages from Kafka's topic to check the latency of the entire pipeline [5]. The test ran continuously for 100 s to check the latency with a change in sampling frequency of interrogator. For each sampling frequency, the test was run for 100 s with replication 2.

4 Proposed methodology

The FBG sensors are characterized by the Bragg wavelength (λ_B) which is sensitive to temperature and strain change. According to the FBGs characteristic sensing principle, the Bragg wavelength shifts with the change in exposed temperature and strain. The FBGs fabrication process is an expensive and highly precise methodology. According to coupled-mode theory, the Bragg wavelength (λ_B) is closely related to the effective refractive index (n_{eff}) and FBG period (Λ) as per Eq. 1 [19, 30].

$$\lambda_B = 2n_{eff}\Lambda \tag{1}$$

In summary, the central reflection wavelength shift ($\Delta\lambda_B$) of the Bragg sensor is affected by temperature (ΔT) and strain (ε), the relationship between these physical parameters is given by the Eq. 2 [15, 19, 30].

$$\frac{\Delta\lambda_B}{\lambda_B} = \left(1 - \rho_e\right)\varepsilon + (\propto + \xi)\Delta T \tag{2}$$

In Eq. 2, the parameters such as ρ_e, $\propto$ and ξ are the photo-elastic coefficients, thermal expansion coefficient, and thermo-optic coefficient respectively. Further, the gauge factor (G) of the FBG strain sensor can also be represented as in Eq. 3.

$$G = \left(1 - \rho_e\right) * \lambda_B \tag{3}$$

Similarly, the reflection and transmission of FBG sensor concerning the wavelength (λ) can be formulated as in Eqs. 4 and 5 respectively.

$$r(\lambda) = \frac{iKsinh(QL)}{Qcosh(QL) - i\Delta\beta sinh(QL)} \tag{4}$$

$$t(\lambda) = \frac{Q}{Qcosh(QL) - i\Delta\beta sinh(QL)} \tag{5}$$

where $Q(\lambda) = \sqrt{|K|^2 - |\Delta\beta|^2}$

The coupling strength is represented by $K(\lambda) = (\Delta n\pi)/\lambda$. Where Δn is the index modulation and L is the grating length. $\Delta\beta = \left(2\pi n_{eff}\right)\left(\lambda^{-1} - {\lambda_B}^{-1}\right)$ is the phase mismatch. The above parameters will allow calculating the transmission and reflection of the FBG sensor. The above equations represent the grating wavelength (λ_B) is closely related to the strain input supplied to the FBGs [1]. Alternatively, to use stress instead of strain (for isotropic material), the relationship between several stress components can be mathematically represented as in Eq. 6.

$$\varepsilon = \frac{1}{E}(\sigma_{zz} + \nu\sigma_{xx} + \nu\sigma_{yy}) \tag{6}$$

where, E is Young's modulus of elasticity and ν is the poison's ratio of the FBG core-cladding material. These are the average value of the FBG core-cladding compound. Similarly, σ_{xx}, σ_{yy} are the transverse stresses and σ_{zz} is the longitudinal stress component [21]. From Eqs. 4 and 5, it is observed that the reflected and transmitted wavelengths from the FBG sensor are just a mirror image to each other [9]. The received power (P) at the photodetector after transmitting twice through a 3 dB coupler, the power can be represented as in Eq. 7. The incident light power per unit wavelength is represented as I_o.

$$P(\lambda) = I_o \int_{-\infty}^{+\infty} r(\lambda)t(\lambda)d\lambda \tag{7}$$

The load calibration model is well tested to evaluate the actual strain using the FBG sensor. There are seven distinct steps such as down sampling, moving average filtering, measurement of peak wavelength, calculation of wavelength shift, load calibration is utilized to predict the actual load applied on the base plate of the suspension bridge model. The configuration settings and analytical modeling behind the above distinct steps are explained in the subsequent subsections.

4.1 Down sampling of the acquired signal

The FBG interrogator is configured to acquire a sensor (1545 nm) signal at 2.5 kHz sampling rate. Therefore, the signal processor must have enough processing capability to gather ample information from the acquired signal. The proposed methodology focuses on developing a simple Digital Signal Processing (DSP) algorithm which can run on low memory, low resource, and low-cost embedded platform. Based on the above-mentioned objective, the down sampling mechanism has been utilized to reduce the signal samples "x(n)" by an integer factor of "M". In the proposed down sampling mechanism, the integer factor "M" is considered as 5. The sampling rate of the original signal is considered as T = 0.4 ms. Hence, the resultant sampling rate after performing down sampling is about T = (0.4 × 5) = 2 ms, which is the same as 0.5 kHz. The reduced signal "y(n)" can be written as mentioned in Eq. 8.

$$y(n) = x(nM) \tag{8}$$

4.2 Moving average filter

The moving average filter is the most commonly adopted and easily developed signal filtering mechanism in almost all the DSP applications. This filtering mechanism is an optimal filtering scheme for reducing random noise while retaining a sharp step response. Due to this advantage, it is widely used in most of the common time-domain filtering schemes but it is having a very bad reputation in frequency domain filtering due to its inability to separate one band of frequencies from another. There are other filtering approaches are available such as Gaussian, Multiple-pass Moving Average, and Blackman filter. These filtering schemes are having better performance in the frequency domain but the computation time is quite high due to its complexity. In this work, a Moving Average Filter of Window Size

of 80 is used and the filtered signal "y(n)" is estimated as shown in Eq. 9 where the input signal is represented as "x(n)".

$$y(n) = (1/\text{window_size})[x(n) + x(n-1) + \cdots + x(n-(\text{window_size}-1))] \quad (9)$$

4.3 Peak wavelength detector and wavelength shift detector

The peak wavelength detector is a procedure to estimate the local as well as absolute maxima and minima from the smoothed signal. The moving average filter is used to smooth the unwanted noisy elements in the acquired signal. The peak detector primarily estimates the global (absolute) maxima at wavelength "x^*" if $f(x^*) \geq f(x)$ for all x in the domain "X". The wavelength corresponding to the peak reflectivity is the reflected wavelength from the FBG sensor. The change in wavelength concerning the base wavelength of the FBG sensor is estimated as the wavelength shift. This wavelength shift is used to calibrate the input load for the corresponding FBG sensor.

4.4 Load calibration model of the FBG sensor

The fundamental of calibration in measurement systems is based on the comparison of measurands by a system under test with those with known accuracy. A Partial Least Square Regression (PLSR) model is developed to estimate the applied load on the top of the base plate by considering the actual wavelength shift. This PLSR model is generally used to model a response variable when there is a large set of predictor variables. PLSR model constructs a new set of predictor variables which is known as components in linear patterns/combinations of the original predictor variables. The PLSR considers the response variables into account; therefore, it often leads to a model with fewer components. The mathematical model to evaluate the applied load on the base plate using the wavelength shift is presented in the results and discussion section of the article. Figure 3 shows the input and output parameters of the proposed PLSR model.

4.5 Kafka for permanent real-time storage and stream analytics

The data in the host computer is processed using LabView and is further sent to the local storage facility for permanent storage and analysis for long term inferences using machine learning and big data principles and tools. As the generated is unbounded streaming data, hence, Kafka has been considered here, a distributed streaming data storage and messaging system, for the storage of real-time data coming from the host computer. Whatsoever, other technologies provide storage and messaging services but have their limitations. However, those technologies are more focused on data delivery guarantees rather than speed and higher throughput. JMS acknowledges each message received which is unnecessary in collecting log data. Each message goes through an entire trip of request "the process-acknowledgement model" which compromises with throughput requirements that are necessary for our scenario. Scribe, developed by Facebook, sends log data over a socket inscribe machines, which in turn periodically aggregates and dumps data periodically in HDFS or NFS. Data Highway project by Yahoo, similarly, aggregates and generate "minute files" which is then dumped into HDFS. It exposes unnecessary details like "minute files". On the other hand, Flume, developed by Cloudera, has distributed support and has extensible

"pipes" and "sinks" which make data very flexible. However, it is primarily used for offline data consumption and based on a push model, where brokers push data to the consumers whereas Kafka follows a pull model where consumers pull data at their maximum capacity without being overloaded.

Kafka works on a publish-subscribe principle where each publisher can publish a set of messages to a Kafka topic which is an individual stream of data given with a topic name. Kafka brokers, a collective set of server computers, then stores the received messages. Kafka consumers are the ones responsible for pulling the data from a particular topic or topics depending on the requirement. Kafka consumers only receive the payload in terms of a stream of bytes. It allows the consumer to choose serialization based on the application requirement. Kafka being a distributed system, it stores messages in multiple partitions and is managed by brokers.

However, some disadvantages like it not being a stream processing system, it can't handle usage surges, and MQTT brokers not supporting high scalability and asynchronous processing are reasons why there is a need for transferring data from MQTT broker to Kafka. The system integration is performed at the laboratory and the Kafka job is tested with real-time FBG sensor signal. The following results were obtained as stated in Table 2. Kafka was operated in default settings.

Kafka runs on JVM, hence, an increase in message size with an increase in sampling frequency causes Kafka to pause longer for garbage collection which causes outliers in latency distribution. In [5], the study suggests that when Kafka reads from the OS cache its latency remains in order of 10 ms; however, when its latency read from disc is around 100 ms. Either way, it's it is good enough in our application.

5 Data communication security issues and solutions

The security aspect is one of the major concerns of IoT systems in any real-time application. Physical security is not the ultimate solution in any IoT mechanisms whereas the software and hardware-based measures are the best practices in any such applications. To achieve optimum security practices, few aspects may be considered to make the internet-based approach more secure. Some of the main such aspects are Device, field gateway, cloud gateway, and secure services. In a similar context, various cloud services are available with functionalities like data processing, data analytics, and management, presentation, and connectivity. In this research work, the implementation is done in.

Table 2 Latency report at a different sampling frequency of FBG sensor interrogation

Sampling frequency (KHz)	P50 (ms)	P99.9 (ms)
0.5	1	9
1	1	15
1.5	1.2	21
2	1.4	26
2.5	1.5	30
3	1.5	34
3.5	1.6	37
4	1.8	40

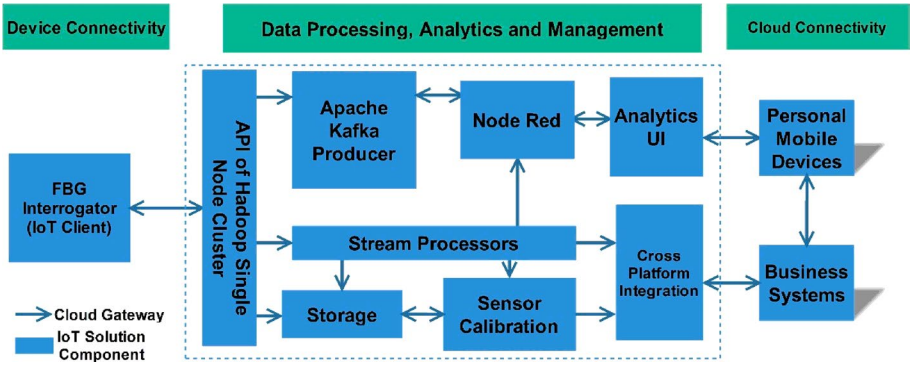
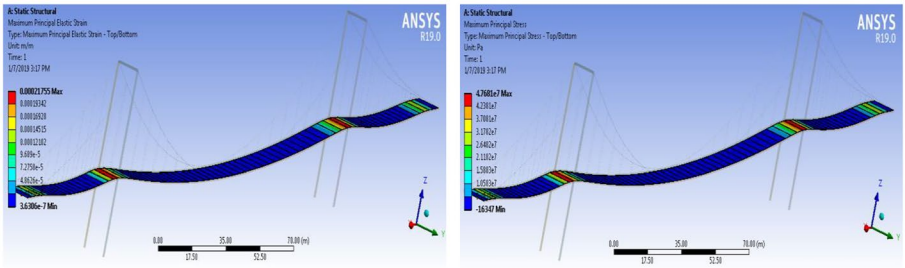


Fig. 4 Proposed data and the communication security model of SHM IoT mechanism



(a) Maximum principal strain distribution

(b) Maximum principal stress distribution

Fig. 5 **a** Maximum principal strain distribution. **b** Maximum principal stress distribution

The proposed data communication scheme is portrayed in Fig. 4 consisting of Device connectivity using FBG Interrogator, Data processing, data analytics and management, and cloud connectivity. The Hadoop single node cluster works as a data processing, analytics, and management unit in the proposed communication scheme. The data processing element represents the sensor calibration and signals processing unit as shown in Fig. 4. The cloud connectivity handles the mobile application and other business integration with the complete real-time data handling scheme.

6 Results and discussion

Firstly, the FEA of the structure is analyzed and the maximum principal strain distribution and maximum principal stress profile are evaluated using ANSYS software. Figure 5a and b shows the strain and stress distribution over the base plate of the structure respectively. It is observed that the maximum principal stress and strain on the based plate is near the two support pillars. FBG sensor is bonded at the center point of the base plate to analyze the mathematical parameters observed from the FEA model and physical signals from the FBG Interrogator.

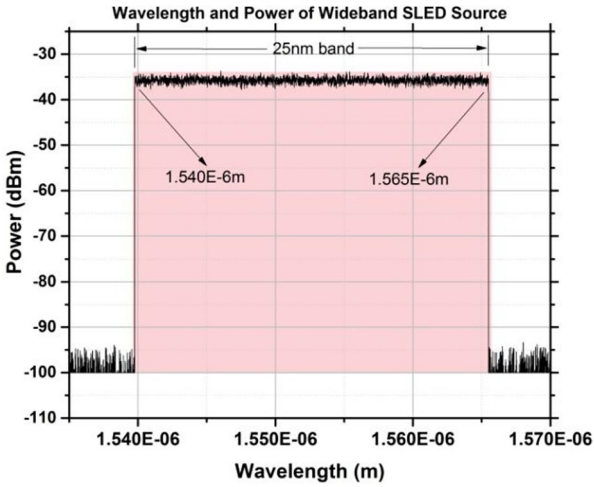


Fig. 6 Response of the wideband SLED source power in dBm

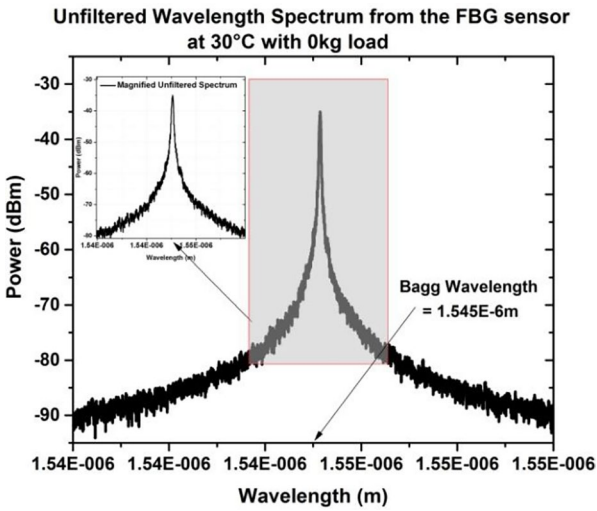


Fig. 7 Reflected wavelength from the FBG sensor at 30 °C with 0 kg load

The further result and analysis portrayed the signal analysis model and evaluation of performance parameters of the same. The primary part of the FBG sensing systems is the tunable laser source. Tunable lasers are widely used as the light sources in the FBG interrogation process for almost all the experiments. In this kind of interrogation system, the tunable sources quickly sweep over a certain range of wavelengths, and reflected light from the FBG sensor is detected by a photo-detector (PD) simultaneously. During the initial configuration of this proposed calibration model, a wideband source is used to produce 1540 nm to 1565 nm light wavelength which is shown in Fig. 6. Further, the reflected wavelength recorded from the FBG sensors having a base wavelength at 1545 nm is shown in Fig. 7.

The peak power of − 34.863 dBm can be observed from the reflected wavelength response. Further, the transmission wavelength of the FBG sensor is evaluated by subtracting the reflected wavelength response from the source wavelength and the response is shown in Fig. 8. It is observed from the Fig. 8 that the power level at 1545 nm is about

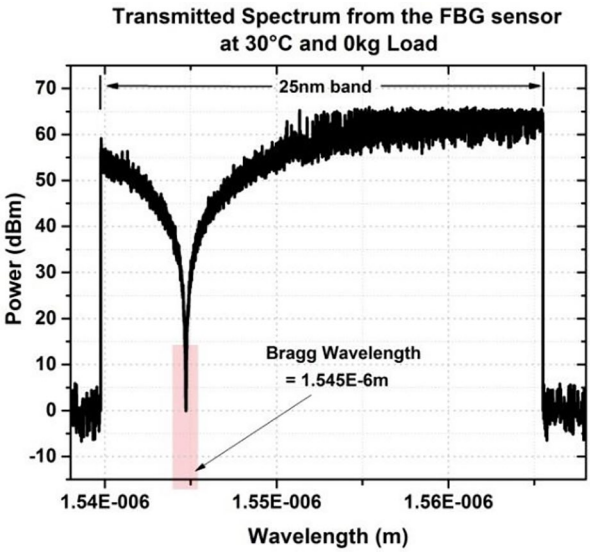


Fig. 8 Transmission wavelength from the FBG sensor without load at 30 °C

− 1.536 dBm (negative) which is quite low. Similarly, it is well understood from the reflection and transmission response that the signal has a lot of noisy samples due to various factors like FBG fiber end interfaces of FBG, index-matching, insertion loss, and bending fiber pigtails, etc. In this article, downsampling and moving average filters are used to reduce the noise level of the sampled signal. The wavelength response after filtering the noise level is shown in Fig. 9. It can be observed from the filtered response that the noise level is quite minimized and the response is smoothed. As the signal amplitude becomes affected after any kind of filtering operation, the maximum wavelength at peak amplitude is evaluated in this work. The peak wavelength is sharply visible at 1545 nm and this effect is observed after the filtering operation of the reflected signal. The proposed peak wavelength detection scheme is utilized to detect the exact value from the filtered response by estimating the global maxima of the response curve.

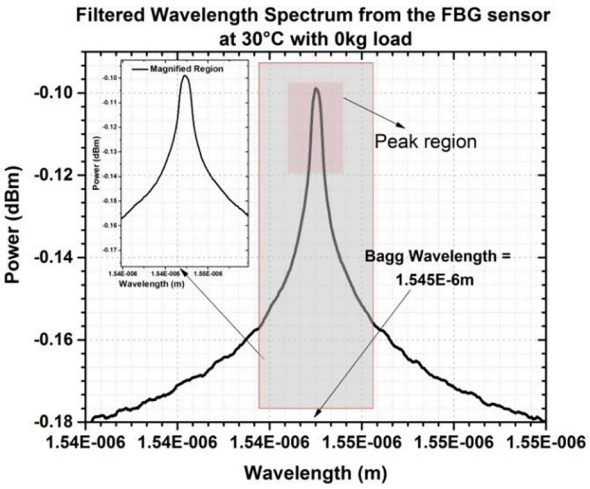


Fig. 9 Reflection wavelength from the FBG sensor after downsampling and filtering

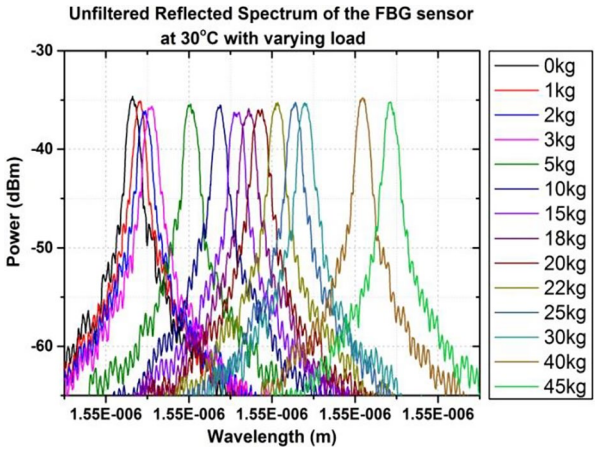


Fig. 10 Unfiltered reflection wavelength from the FBG sensor at 30 °C with varying load

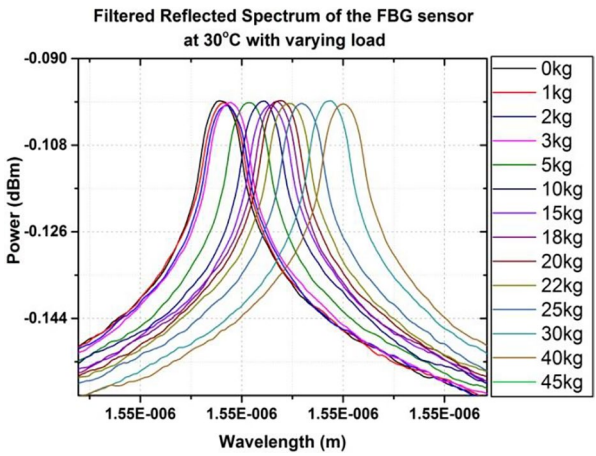


Fig. 11 Filtered reflection wavelength from the FBG sensor at 30 °C with varying load

Similarly, the unfiltered reflection wavelength shift from the FBG sensor at 30 °C and varying applied load is shown in Fig. 10. In a similar context, the filtered reflection wavelength shift from the FBG sensor at 30 °C and varying applied load is shown in Fig. 11. It is observed that the noisy signal is efficiently filtered out using the proposed filtering scheme. The peak amplitude is better visible and the exact maxima can easily be calculated.

Furthermore, the relationship between peak wavelength shifts with varying load is shown in Fig. 12, where the graph is quite linear. The PLSR model is used to build a mathematical relationship between wavelength shift and applied load in kg which is shown in Fig. 13. The residual error of the PLSR calibration model for the different applied loads is shown in Fig. 14. It is observed from the Fig. 14 that the error band limits between − 0.9 to 0.2 in the amplitude level. The error band is quite low and the PLSR model fits the target load magnitudes. The percentage of maximum deviation of the proposed PLSR model is about − 2% to 0.4%. The percentage of maximum deviation is quite low which does not affect the performance of measurement. The PLSR model is developed with MSE = 0.0722, RMSE = 0.2688, R-Squared value = 0.9999. The mathematical model to

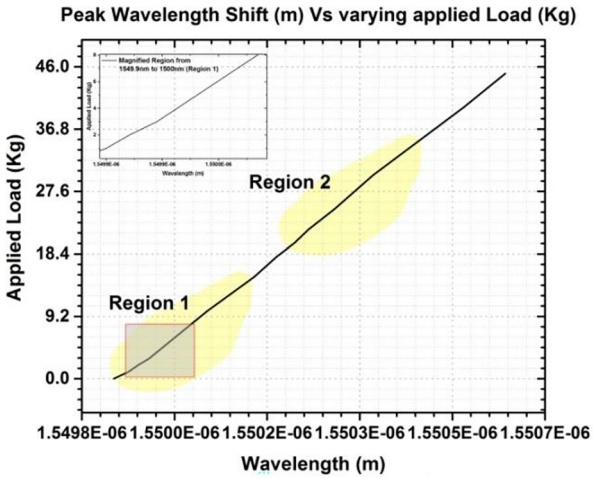


Fig. 12 Relationship between peak wavelength shift Vs varying input load

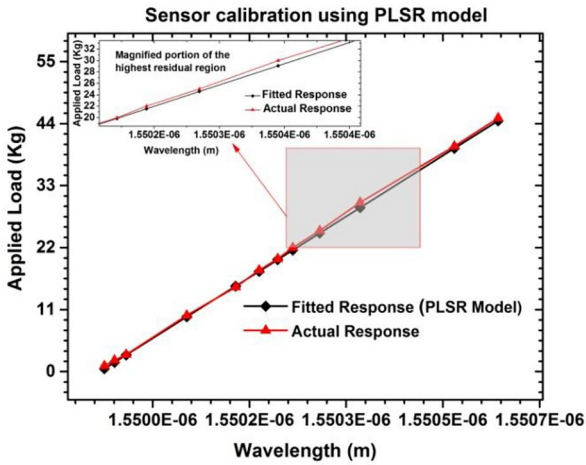


Fig. 13 PLSR calibration model for varying input load

estimate the applied load on the base plate is shown in Eq. 10. Here λ is the shifted wavelength after applying varying load on the base plate of the suspension bridge model.

$$\text{Load} = [-9.3045598\text{e}4] + [(6.0034\text{e}10) * \lambda] \tag{10}$$

Finally, a Graphical User Interface (GUI) is developed to visualize the applied load on the base plate of the suspension bridge which is shown in Fig. 15. The GUI framework shows the actual wavelength shift, the temperature of the base plate near the FBG sensor, the voltage output of the loadcell after the signal conditioning circuit. The LabVIEW application is linked with the Hadoop single node cluster using the TCP/IP address of the server. The loadcell was installed to evaluate a comparative analysis with the FBG sensing mechanism. The PLSR model is used to estimate the load applied on the base plate of the suspension bridge model.

Finally, the Decision Support model is implemented in the GUI framework to generate adequate notifications. Few test notifications are listed in Table 3. The proposed decision

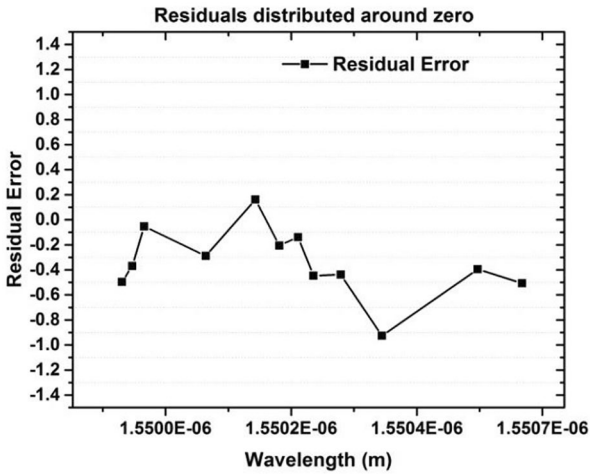


Fig. 14 Residual error of the PLSR calibration model for the different applied load

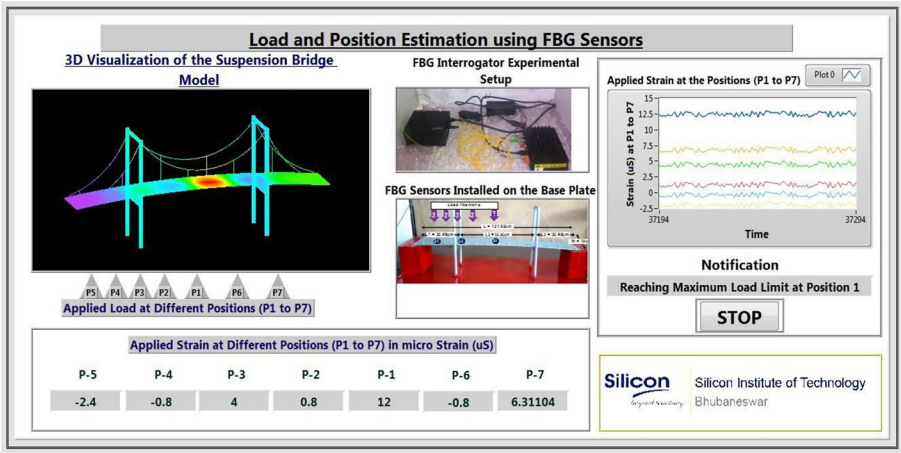


Fig. 15 GUI to visualize the applied load on the base plate of the suspension bridge

Table 3 Test results of the developed GUI application

Sl. no.	Applied load (Kg)	Notification
1	0	No load applied on the base plate
2	23	Mid threshold limit exceeds
3	44	Reaching maximum load limit

model to generate adequate notifications is shown in Fig. 16. The FBG sensor was bonded with standard bonding methodology and the corresponding signal analysis is performed.

A scalable architecture of the proposed FBG based distributed structural health monitoring mechanism is shown in the Fig. 17. The scalable architecture of the distributed sensing

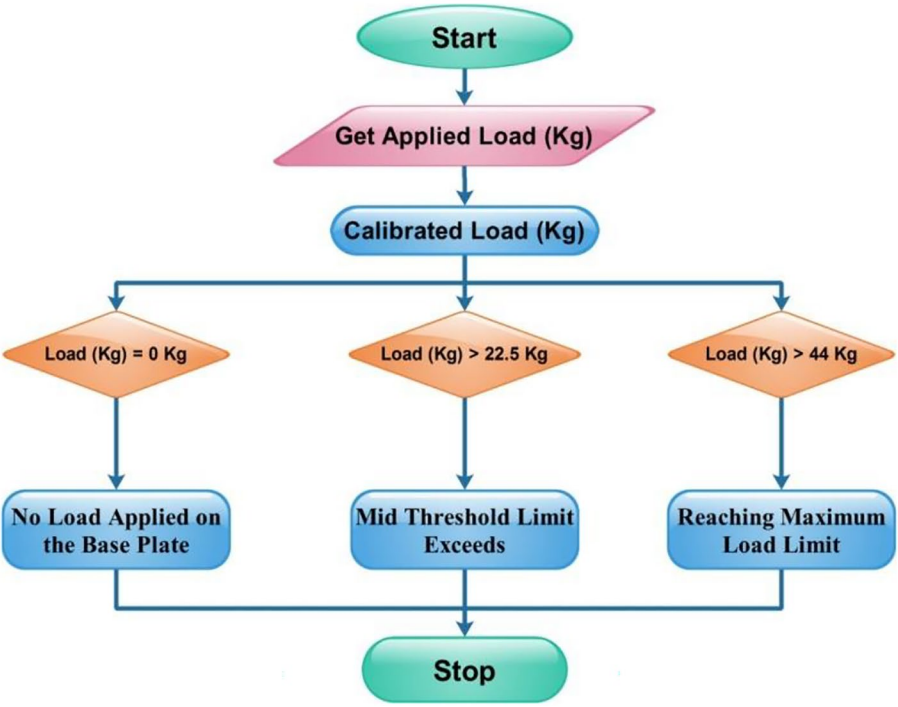


Fig. 16 Decision support model to generate adequate notifications

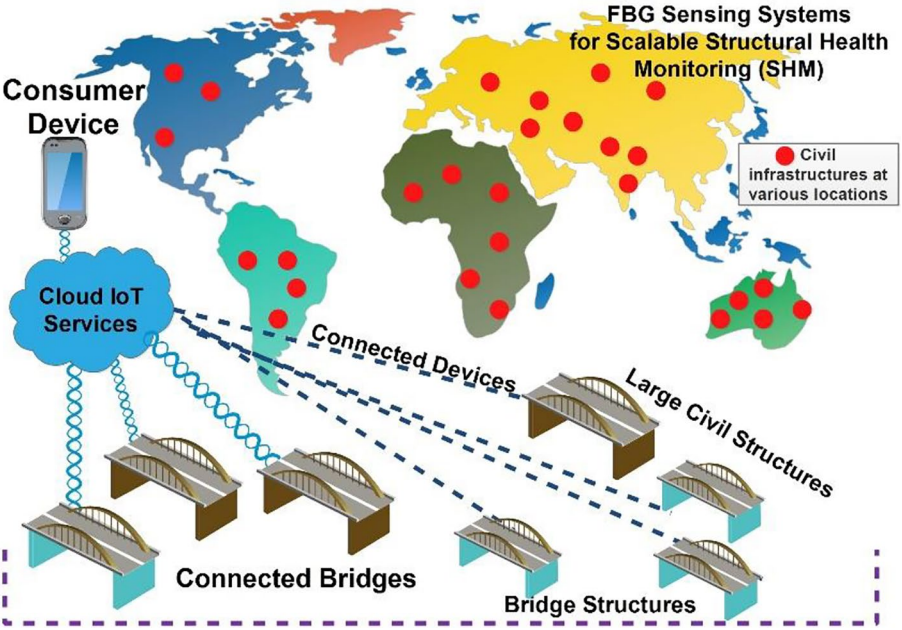


Fig. 17 Scalable architecture of the distributed structural health monitoring system

has multiple civil structures connected to a cloud IoT service. The consumer devices represent the user devices through which the real-time data visualization can be performed.

7 Conclusions

FBG sensing technology is widely accepted in many SHM applications. The IoT enabled FBG sensing solution is one of the promising technological requirements for high-speed measurement. The integration of IoT enabled FBG sensing technology and DSS makes the overall system smart enough to provide automated decisions. However, the calibration of the FBG sensor is one of the most challenging aspects of various real-time applications of FBG sensors. Efficient signal processing and calibration models are required to extract useful information from the noisy signal. The FBG sensor is calibrated using the proposed signal processing and detection schemes. The proposed calibration model is efficiently evaluated by bonding one FBG sensor on the base plate of a suspension bridge model. The experimental validation of the calibration model is performed at the laboratory environment. The static load calibration is successfully implemented during this research work and errors are recorded. The PLSR based load estimation model produces almost similar predictors with an R-Squared value of 0.9999 and MSE of 0.0722. Finally, the PLSR model of the static load calibration scheme is implemented in the GUI application along with the DSS model to visualize every aspect of the sensing mechanism during real-time testing.

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